Первое высшее техническое учебное заведение России
Санкт-Петербургский горный университет императрицы Екатерины II



Основы программирования измерительных систем

Лекция 3. RGB светодиод.

Функции и рефакторинг кода

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2026

ГОРНЫЙ

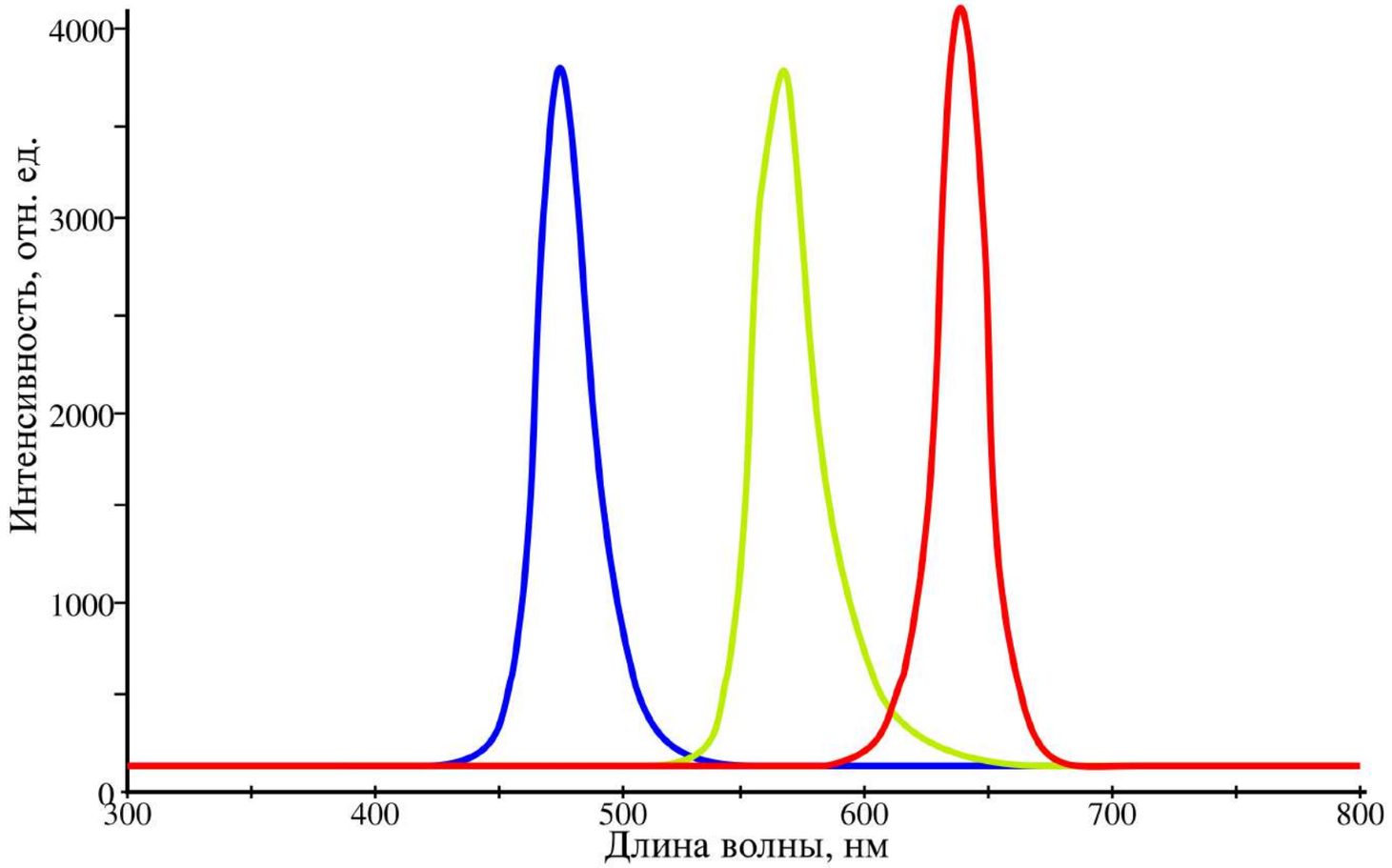
RGB светодиод





ГОРНЫЙ

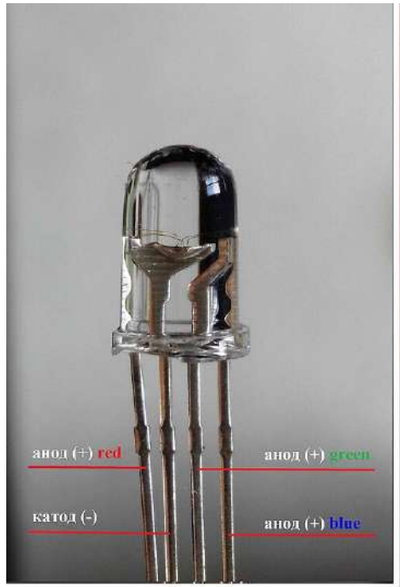
RGB светодиод





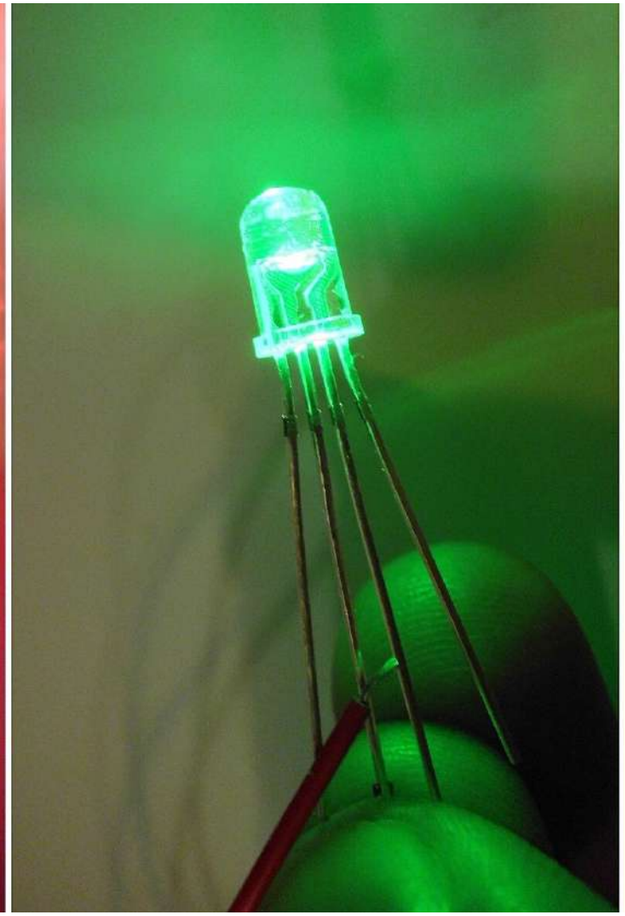
ГОРНЫЙ

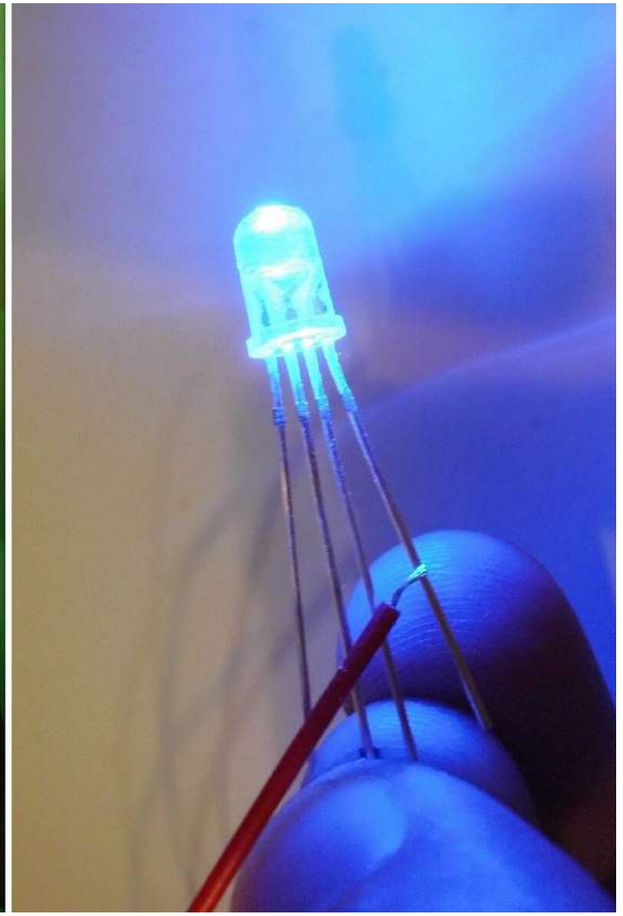
RGB светодиод





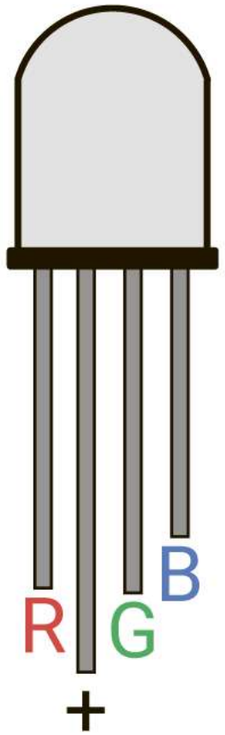


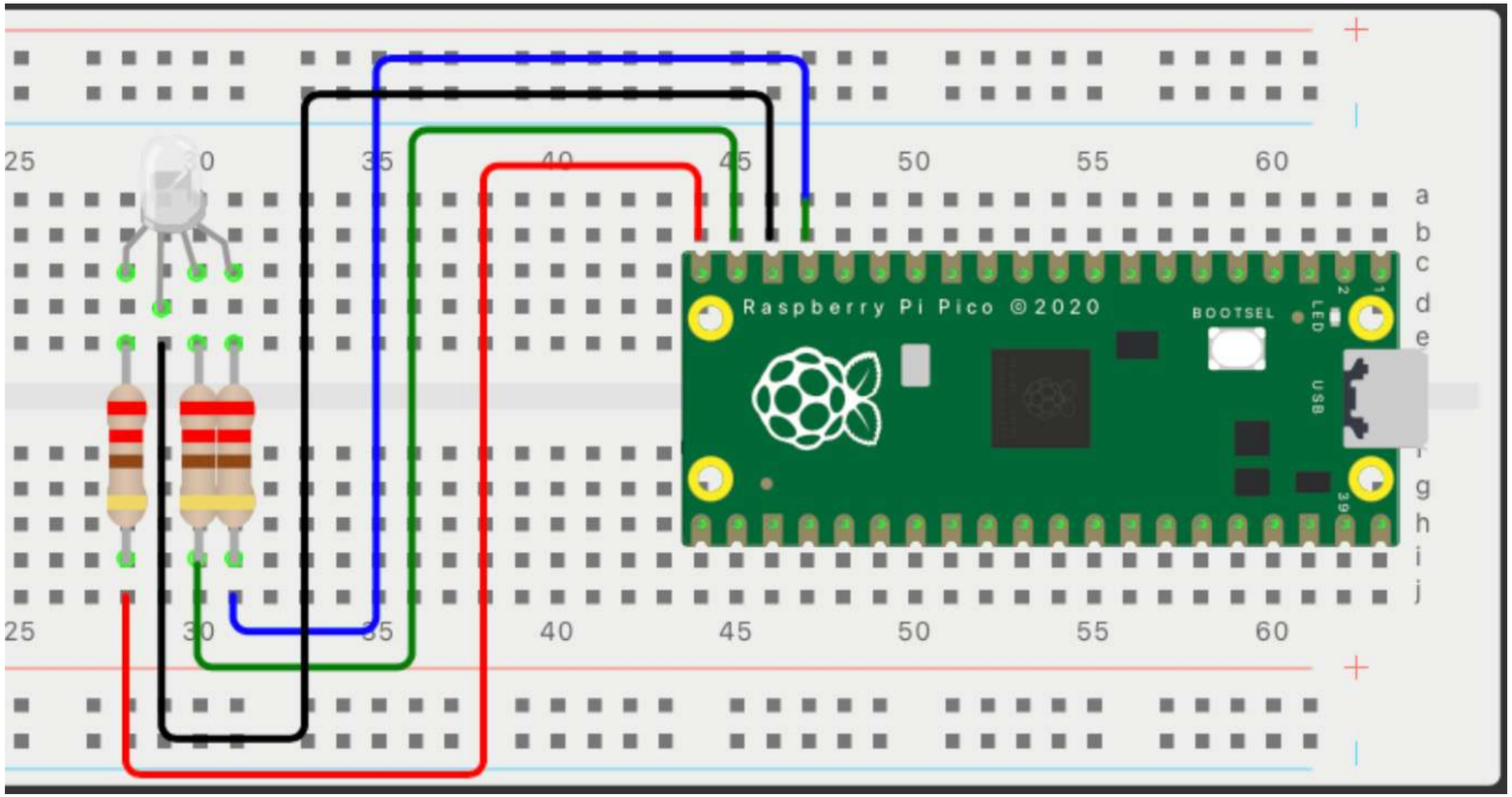




ГОРНЫЙ

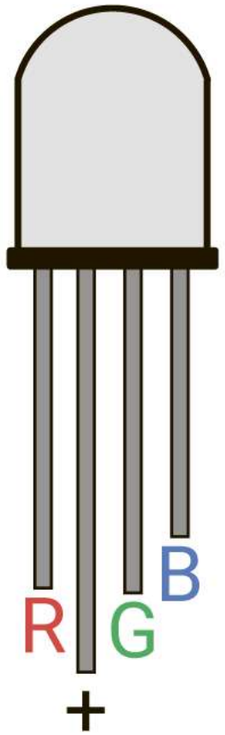
Подключение RGB светодиода

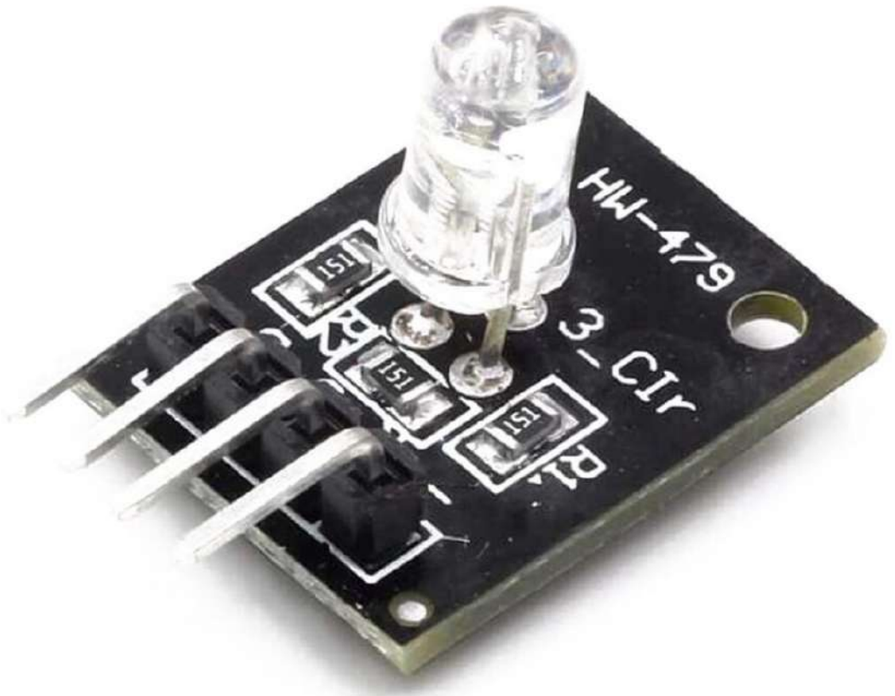




ГОРНЫЙ

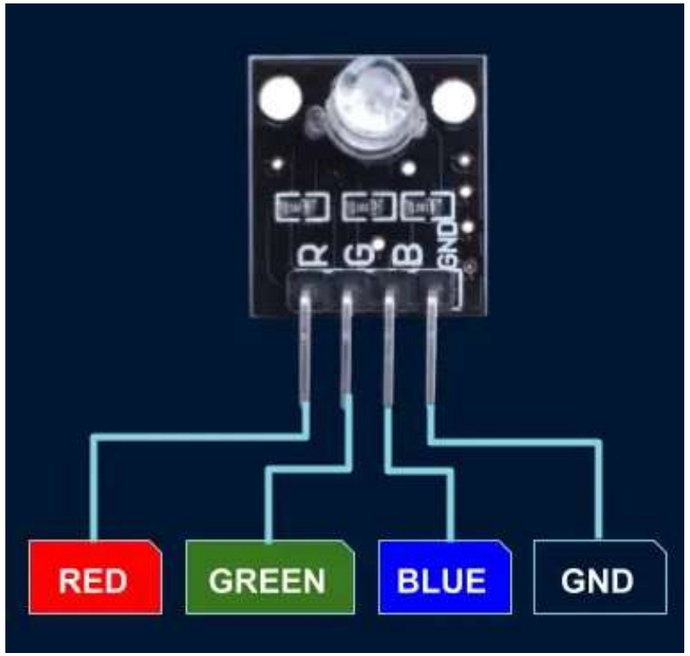
RGB модуль

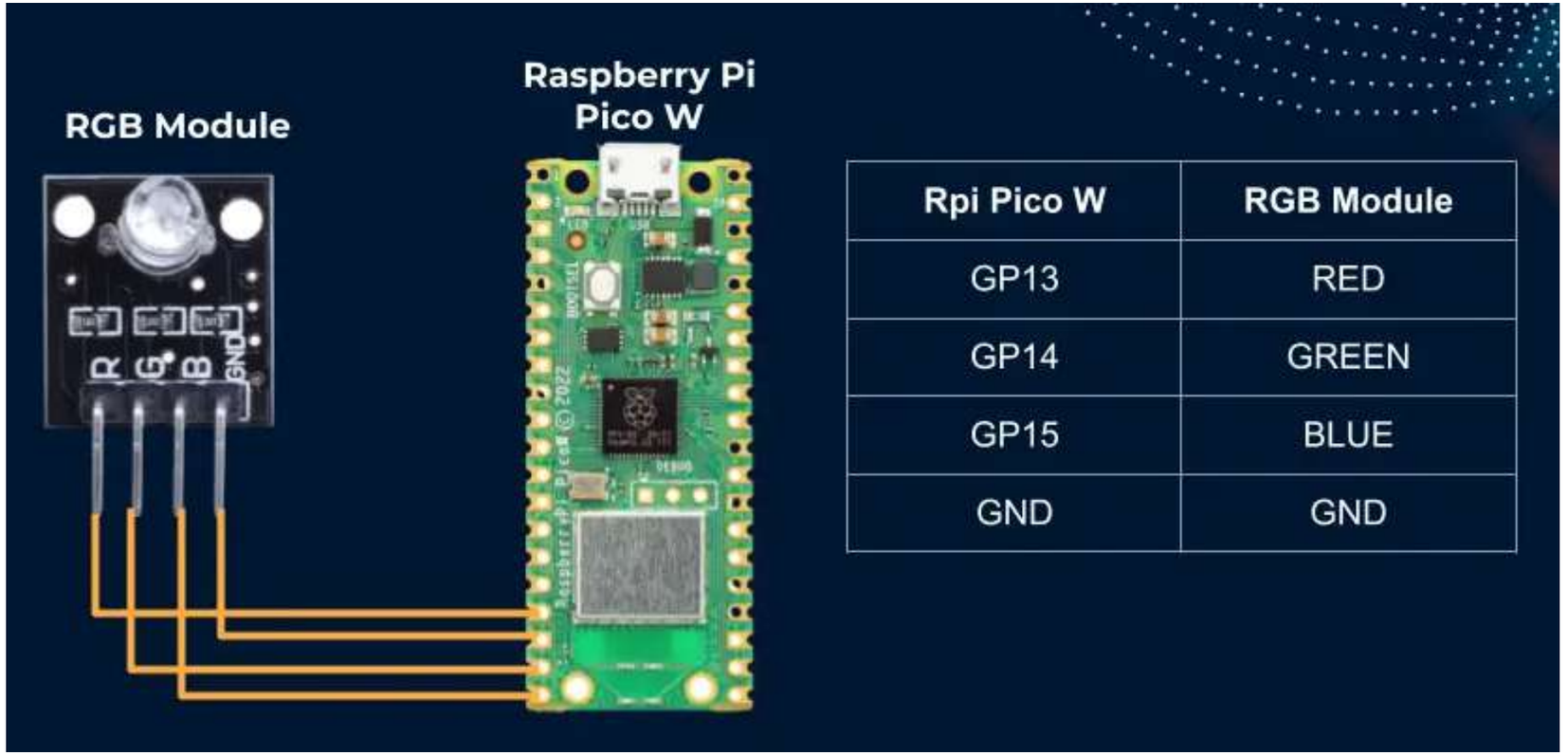




ГОРНЫЙ

RGB модуль

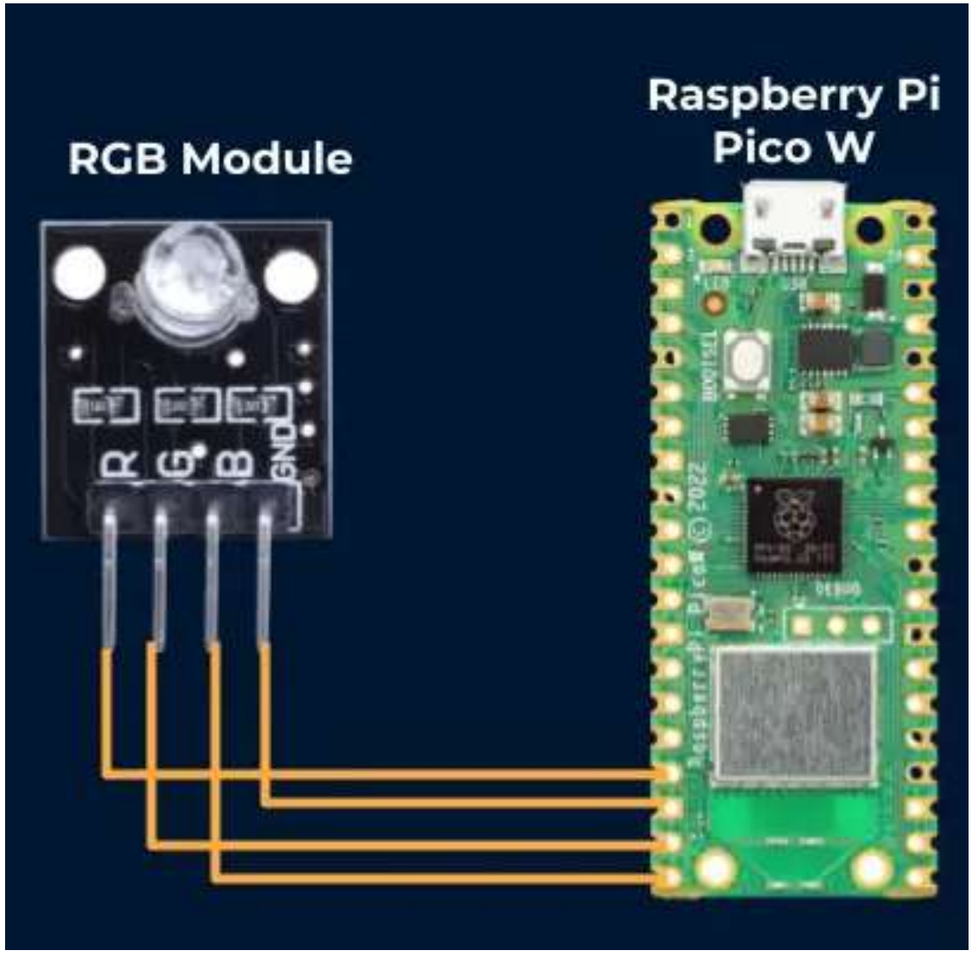




Rpi Pico W	RGB Module
GP13	RED
GP14	GREEN
GP15	BLUE
GND	GND

ГОРНЫЙ

RGB модуль






```python
from machine import Pin
from time import sleep

RED_PIN = 13
GREEN_PIN = 14
BLUE_PIN = 15

red_led = Pin(RED_PIN, Pin.OUT)
green_led = Pin(GREEN_PIN, Pin.OUT)
blue_led = Pin(BLUE_PIN, Pin.OUT)

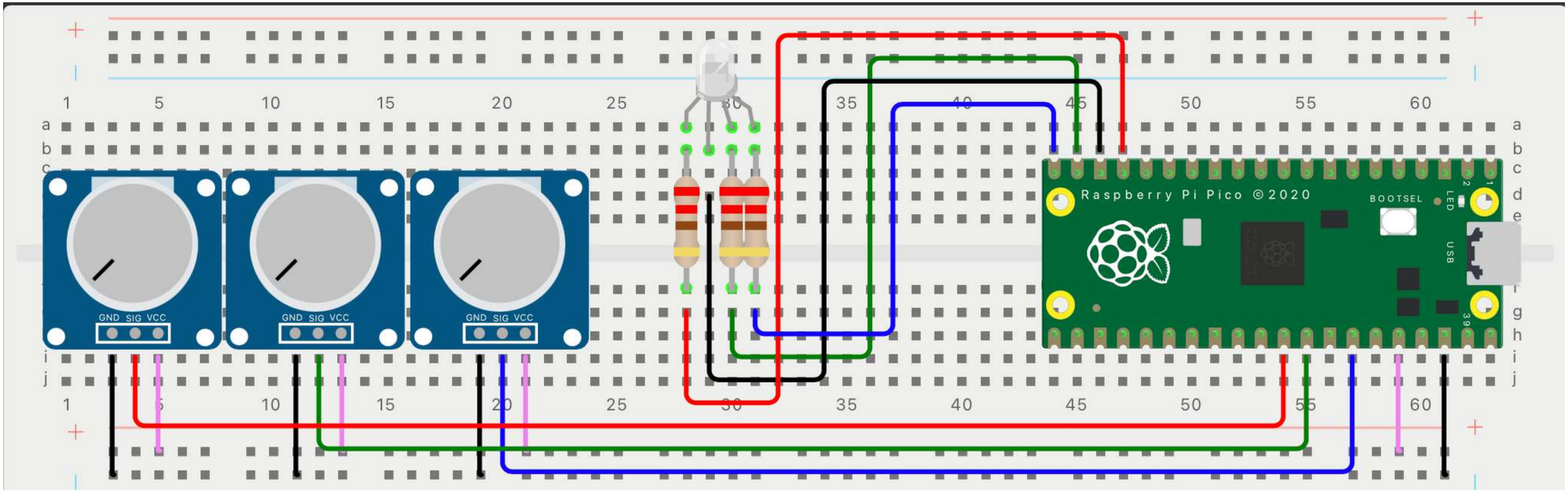
while True:
    red_led.on()
    green_led.off()
    blue_led.off()
    sleep(1)

    red_led.off()
    green_led.on()
    blue_led.off()
    sleep(1)

    red_led.off()
    green_led.off()
    blue_led.on()
    sleep(1)
```

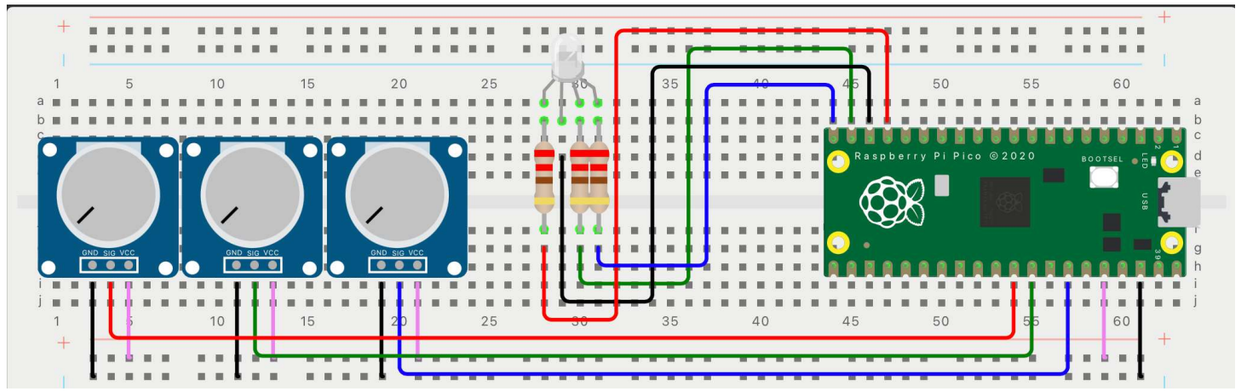
ГОРНЫЙ

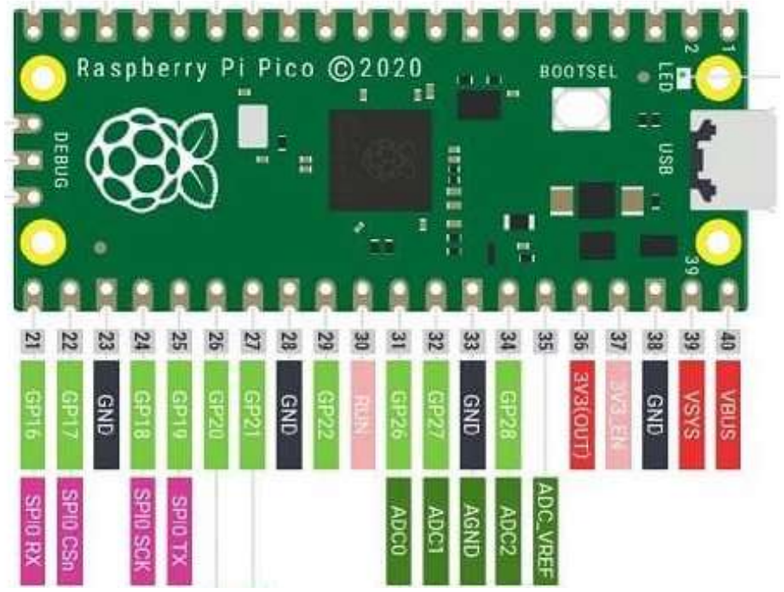
RGB модуль



ГОРНЫЙ

RGB модуль






```
from machine import Pin, ADC, PWM
from time import sleep


RED_PIN = 13
GREEN_PIN = 14
BLUE_PIN = 15

RED_POT = 26
GREEN_POT = 27
BLUE_POT = 28

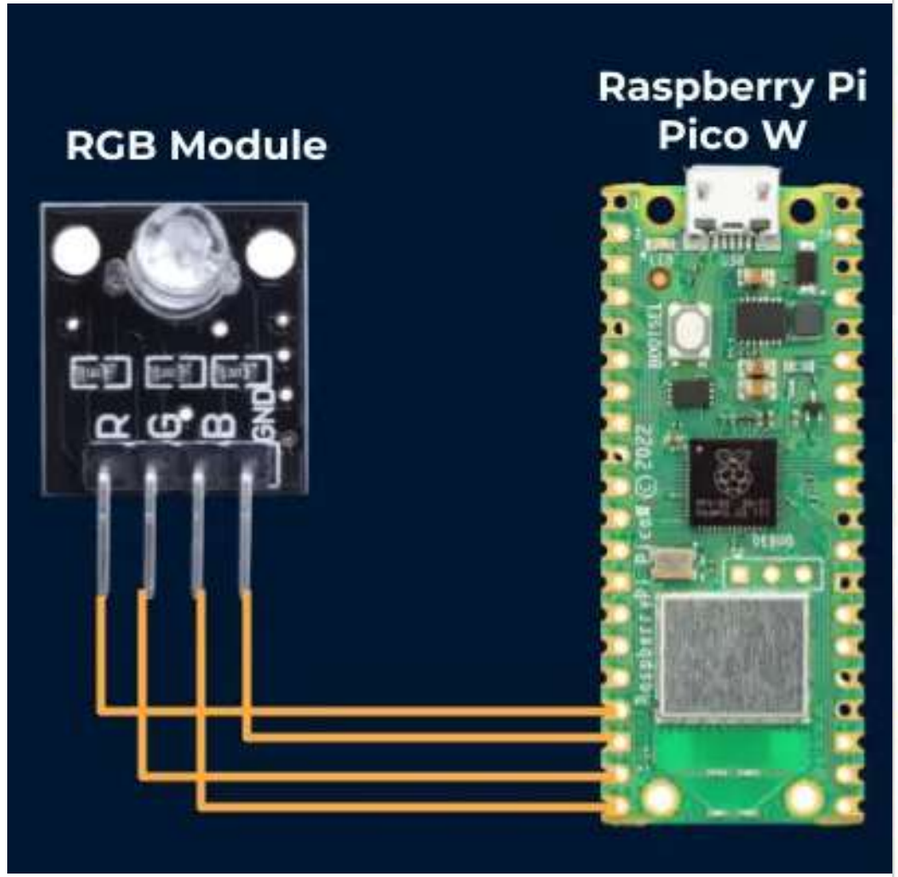
red_led = PWM(RED_PIN, freq=1000)
green_led= PWM(GREEN_PIN, freq=1000)
blue_led = PWM(BLUE_PIN, freq=1000)

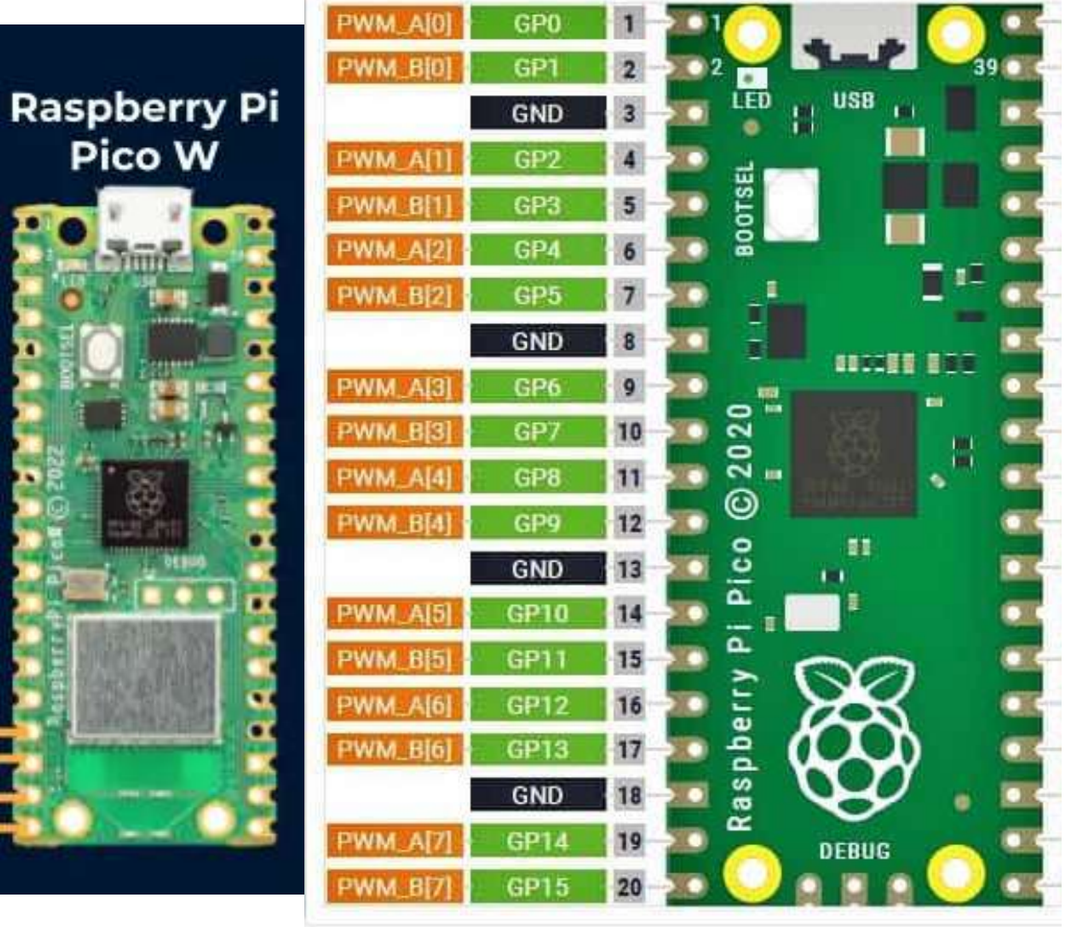
red_pot = ADC(Pin(RED_POT))
green_pot = ADC(Pin(GREEN_POT))
blue_pot = ADC(Pin(BLUE_POT))


while True:
    red_led.duty_u16(red_pot.read_u16())
    green_led.duty_u16(green_pot.read_u16())
    blue_led.duty_u16(blue_pot.read_u16())
    sleep(0.001)
```

ГОРНЫЙ

RGB модуль






```
from machine import PWM
from time import sleep

RED_PIN = 13
GREEN_PIN = 14
BLUE_PIN = 15

red_led = PWM(RED_PIN, freq=1000)
green_led= PWM(GREEN_PIN, freq=1000)
blue_led = PWM(BLUE_PIN, freq=1000)

while True:
    red_led.duty_u16(65535)
    green_led.duty_u16(0)
    blue_led.duty_u16(0)
    sleep(1)
    red_led.duty_u16(0)
    green_led.duty_u16(65535)
    blue_led.duty_u16(0)
    sleep(1)
    red_led.duty_u16(0)
    green_led.duty_u16(0)
    blue_led.duty_u16(65535)
    sleep(1)
```

ГОРНЫЙ

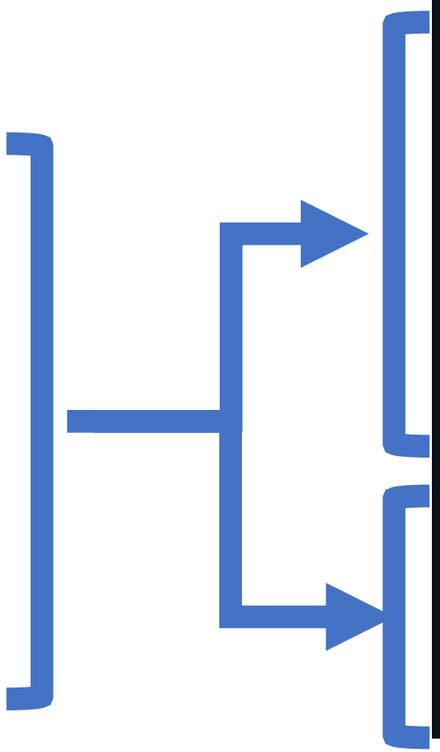
RGB модуль

```
from machine import PWM
from time import sleep

RED_PIN = 13
GREEN_PIN = 14
BLUE_PIN = 15

red_led = PWM(RED_PIN, freq=1000)
green_led= PWM(GREEN_PIN, freq=1000)
blue_led = PWM(BLUE_PIN, freq=1000)

while True:
    red_led.duty_u16(65535)
    green_led.duty_u16(0)
    blue_led.duty_u16(0)
    sleep(1)
    red_led.duty_u16(0)
    green_led.duty_u16(65535)
    blue_led.duty_u16(0)
    sleep(1)
    red_led.duty_u16(0)
    green_led.duty_u16(0)
    blue_led.duty_u16(65535)
    sleep(1)
```



```
from machine import PWM
from time import sleep

RED_PIN = 13
GREEN_PIN = 14
BLUE_PIN = 15

red_led = PWM(RED_PIN, freq=1000)
green_led= PWM(GREEN_PIN, freq=1000)
blue_led = PWM(BLUE_PIN, freq=1000)

red = (65535, 0, 0)
green = (0, 65535, 0)
blue = (0, 0, 65535)

magenta = (65535, 0, 65535)
yellow = (65535, 65535, 0)
cian = (0, 65535, 65535)
orange = (65535, 32768, 0)
white = (65535, 65535, 65535)

while True:
    for color in red, orange, yellow, green, cian, blue, magenta:
        red_led.duty_u16(color[0])
        green_led.duty_u16(color[1])
        blue_led.duty_u16(color[2])
        sleep(1)
```

ГОРНЫЙ

RGB модуль

```
from machine import PWM
from time import sleep

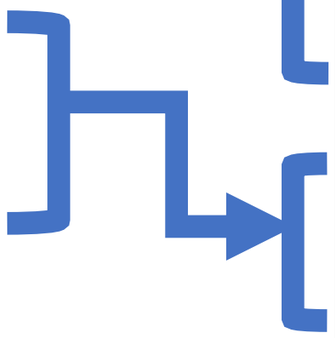
RED_PIN = 13
GREEN_PIN = 14
BLUE_PIN = 15

red_led = PWM(RED_PIN, freq=1000)
green_led= PWM(GREEN_PIN, freq=1000)
blue_led = PWM(BLUE_PIN, freq=1000)

red = (65535, 0, 0)
green = (0, 65535, 0)
blue = (0, 0, 65535)

magenta = (65535, 0, 65535)
yellow = (65535, 65535, 0)
cian = (0, 65535, 65535)
orange = (65535, 32768, 0)
white = (65535, 65535, 65535)

while True:
    for color in red, orange, yellow, green, cian, blue, magenta:
        red_led.duty_u16(color[0])
        green_led.duty_u16(color[1])
        blue_led.duty_u16(color[2])
        sleep(1)
```



```
from machine import PWM
from time import sleep

RED_PIN = 15
GREEN_PIN = 14
BLUE_PIN = 13

red_led = PWM(RED_PIN, freq=1000)
green_led= PWM(GREEN_PIN, freq=1000)
blue_led = PWM(BLUE_PIN, freq=1000)

rgb_led = (red_led, green_led, blue_led)

red = (255, 0, 0)
green = (0, 255, 0)
blue = (0, 0, 255)

magenta = (255, 0, 255)
yellow = (255, 255, 0)
cian = (0, 255, 255)
orange = (255, 128, 0)
white = (255, 255, 255)


def switch_on_a_colour_of_the_specified_brightness(rgb_led, color_tuple,
brightness=1):
    for idx, color in enumerate(color_tuple):
        rgb_led[idx].duty_u16(int(color * 65535 / 255 * brightness))


while True:
    for color in red, orange, yellow, green, cian, blue, magenta:
        switch_on_a_colour_of_the_specified_brightness(rgb_led, color)
        sleep(1)
```

RGB модуль

```python
from machine import PWM
from time import sleep

RED_PIN = 15
GREEN_PIN = 14
BLUE_PIN = 13

red_led = PWM(RED_PIN, freq=1000)
green_led= PWM(GREEN_PIN, freq=1000)
blue_led = PWM(BLUE_PIN, freq=1000)

rgb_led = (red_led, green_led, blue_led)

red = (255, 0, 0)
green = (0, 255, 0)
blue = (0, 0, 255)

magenta = (255, 0, 255)
yellow = (255, 255, 0)
cian = (0, 255, 255)
orange = (255, 128, 0)
white = (255, 255, 255)


def switch_on_a_colour_of_the_specified_brightness(rgb_led, color_tuple, brightness=1):
    for idx, color in enumerate(color_tuple):
        rgb_led[idx].duty_u16(int(color * 65535 / 255 * brightness))


while True:
    for color in red, orange, yellow, green, cian, blue, magenta:
        switch_on_a_colour_of_the_specified_brightness(rgb_led, color)
        sleep(1)
```

```python
def get_colour_data_for_pwm(color_8bit_tuple):
    color_16bit_tuple = tuple([int(rgb * 65535 / 255) for rgb in color_8bit_tuple])
    return color_16bit_tuple


def set_the_colour_brightness(color_tuple, brightness):
    colour_with_a_specified_brightness = tuple([int(rgb * brightness / 100) for rgb in color_tuple])
    return colour_with_a_specified_brightness


def switch_on_a_color(rgb_led, color_tuple):
    for idx, color in enumerate(color_tuple):
        rgb_led[idx].duty_u16(color)


def switch_on_a_colour_of_the_specified_brightness(rgb_led, color_tuple, brightness=1):
    color_tuple = get_colour_data_for_pwm(color_tuple)
    color_tuple = set_the_colour_brightness(color_tuple, brightness)
    switch_on_a_color(rgb_led, color_tuple)


while True:
    for color in red, orange, yellow, green, cian, blue, magenta:
        for brightness in range(100):
            switch_on_a_colour_of_the_specified_brightness(rgb_led, color, brightness)
            sleep(0.01)
        for brightness in range(100, 0, -1):
            switch_on_a_colour_of_the_specified_brightness(rgb_led, color, brightness)
            sleep(0.01)
```

RGB модуль

```python
from machine import PWM
from time import sleep

RED_PIN = 15
GREEN_PIN = 14
BLUE_PIN = 13

red_led = PWM(RED_PIN, freq=1000)
green_led= PWM(GREEN_PIN, freq=1000)
blue_led = PWM(BLUE_PIN, freq=1000)

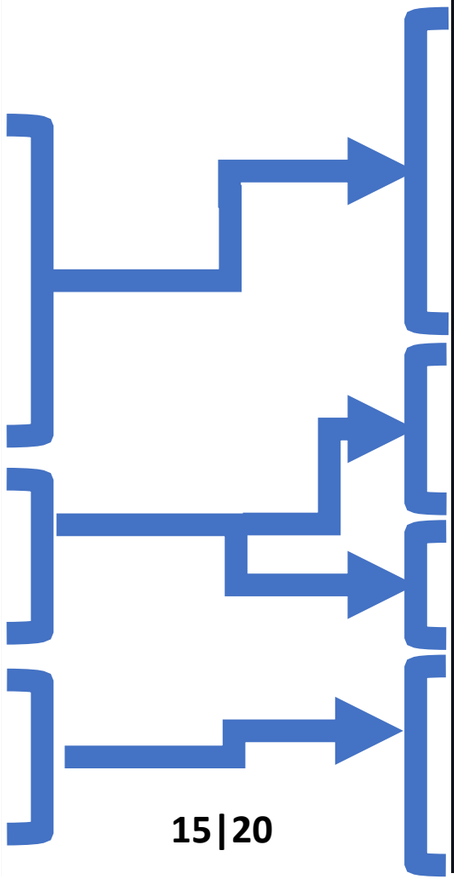
rgb_led = (red_led, green_led, blue_led)

red = (255, 0, 0)
green = (0, 255, 0)
blue = (0, 0, 255)

magenta = (255, 0, 255)
yellow = (255, 255, 0)
cian = (0, 255, 255)
orange = (255, 128, 0)
white = (255, 255, 255)


def switch_on_a_colour_of_the_specified_brightness(rgb_led, color_tuple,
brightness=1):
    for idx, color in enumerate(color_tuple):
        rgb_led[idx].duty_u16(int(color * 65535 / 255 * brightness))


while True:
    for color in red, orange, yellow, green, cian, blue, magenta:
        switch_on_a_colour_of_the_specified_brightness(rgb_led, color)
        sleep(1)
```



```python
from machine import PWM
from time import sleep

RED_PIN = 15
GREEN_PIN = 14
BLUE_PIN = 13

red_led = PWM(RED_PIN, freq=1000)
green_led = PWM(GREEN_PIN, freq=1000)
blue_led = PWM(BLUE_PIN, freq=1000)

rgb_led = (red_led, green_led, blue_led)

colors = {"red": (255, 0, 0),
          "green": (0, 255, 0),
          "blue": (0, 0, 255),
          "magenta": (255, 0, 255),
          "yellow": (255, 255, 0),
          "cian": (0, 255, 255),
          "orange": (255, 128, 0),
          "white": (255, 255, 255),
          "black": (0, 0, 0),
          }

def get_colour_data_for_pwm(color_8bit_tuple):
    color_16bit_tuple = tuple([int(rgb * 65535 / 255) for rgb in
color_8bit_tuple])
    return color_16bit_tuple

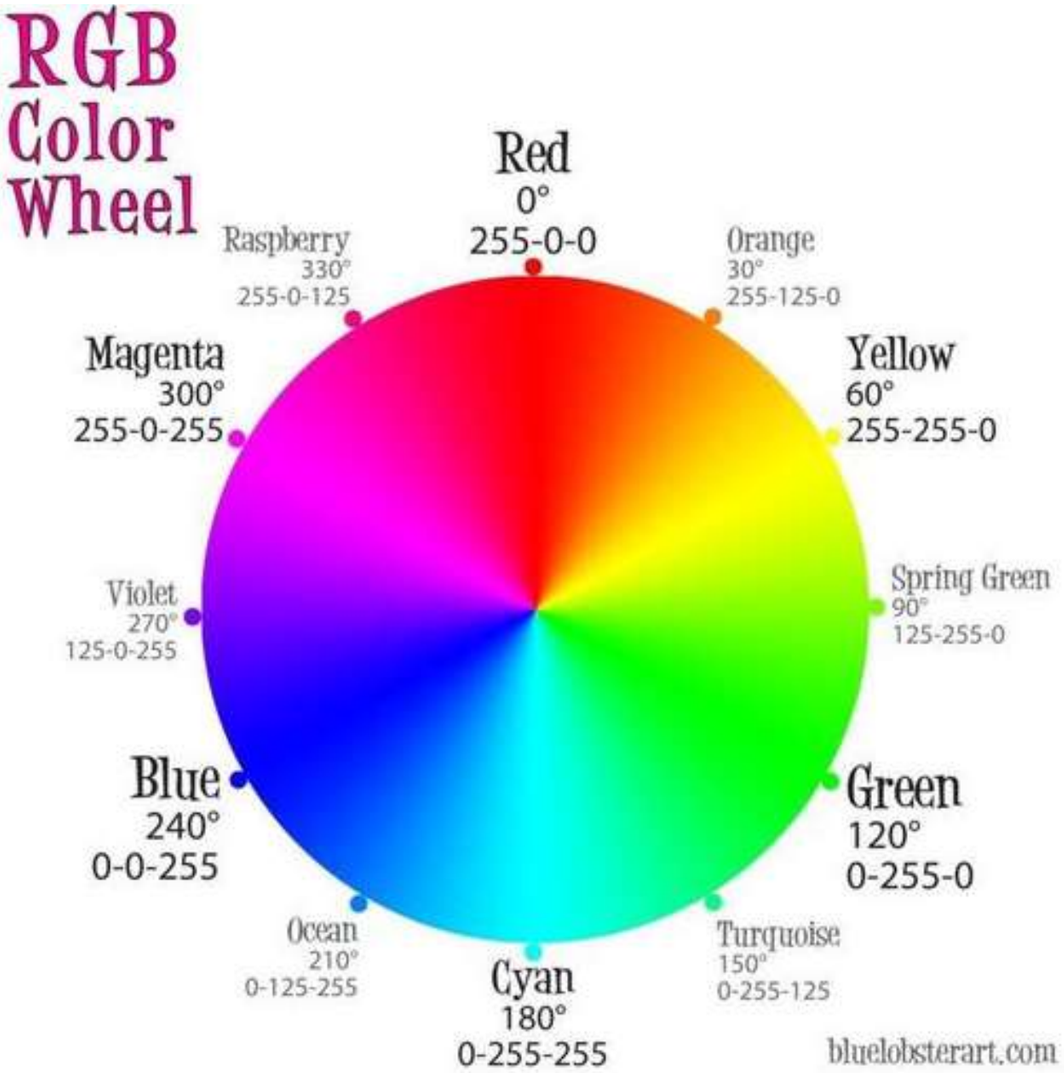
def switch_on_a_color(rgb_led, color_tuple):
    for idx, color in enumerate(color_tuple):
        rgb_led[idx].duty_u16(color)

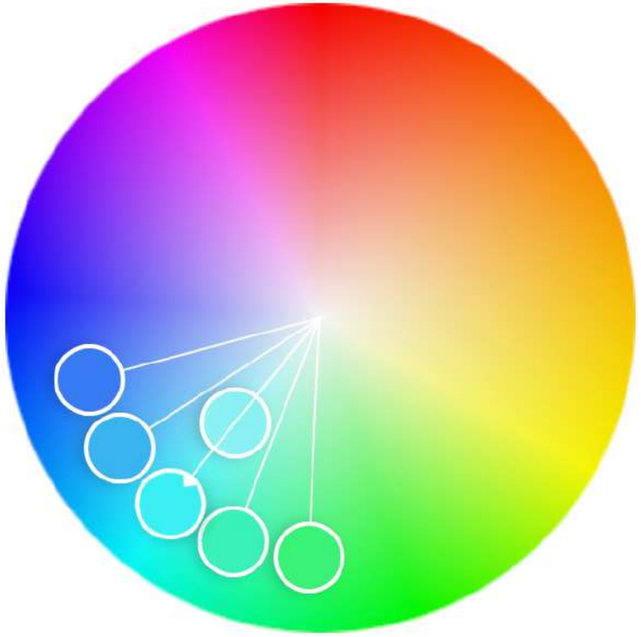
print(*colors.keys())
while True:
    for color, color_tuple in colors.items():
        switch_on_a_color(rgb_led, get_colour_data_for_pwm(color_tuple))
        print(color + "        ", end="\r")
        sleep(1)
```


Обратных ссылок: 0

ГОРНЫЙ

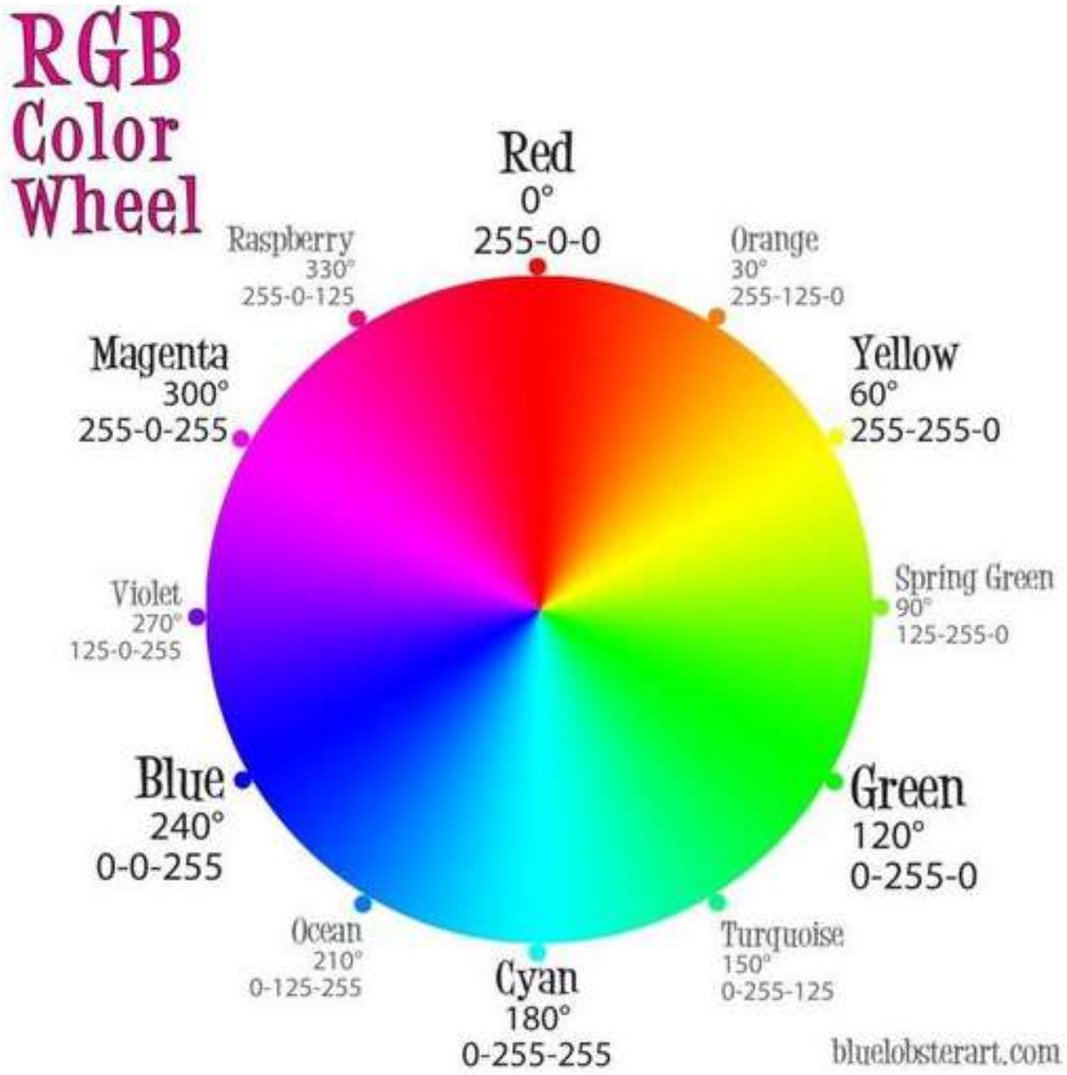
Цветовой круг





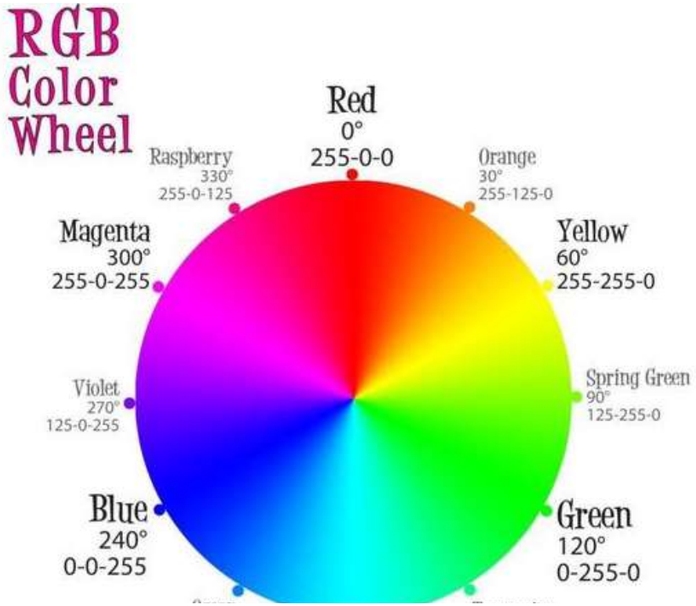
ГОРНЫЙ

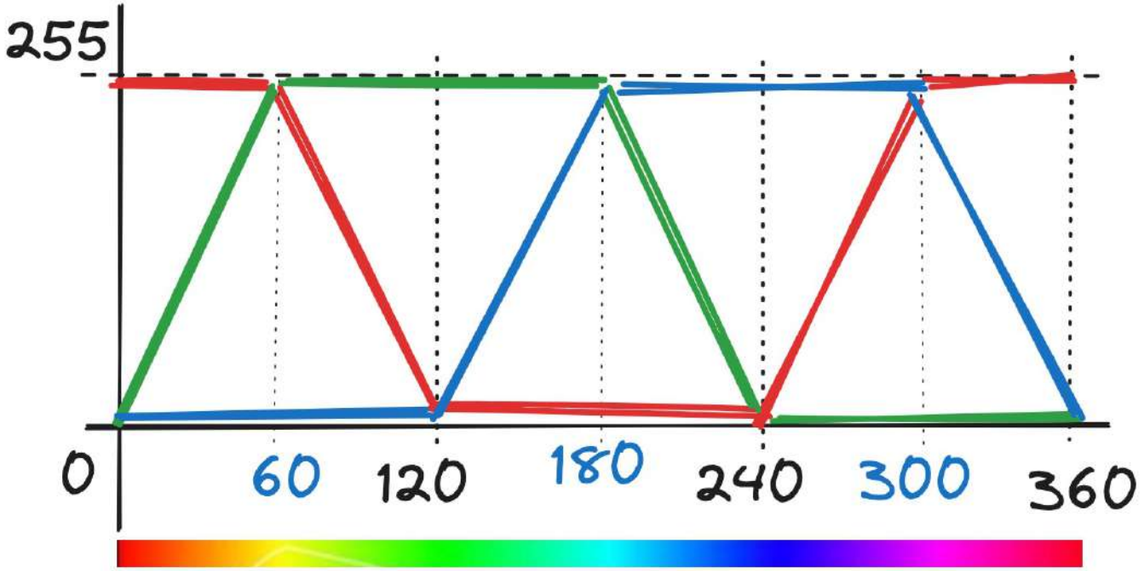
Цветовой круг



ГОРНЫЙ

Цветовой круг

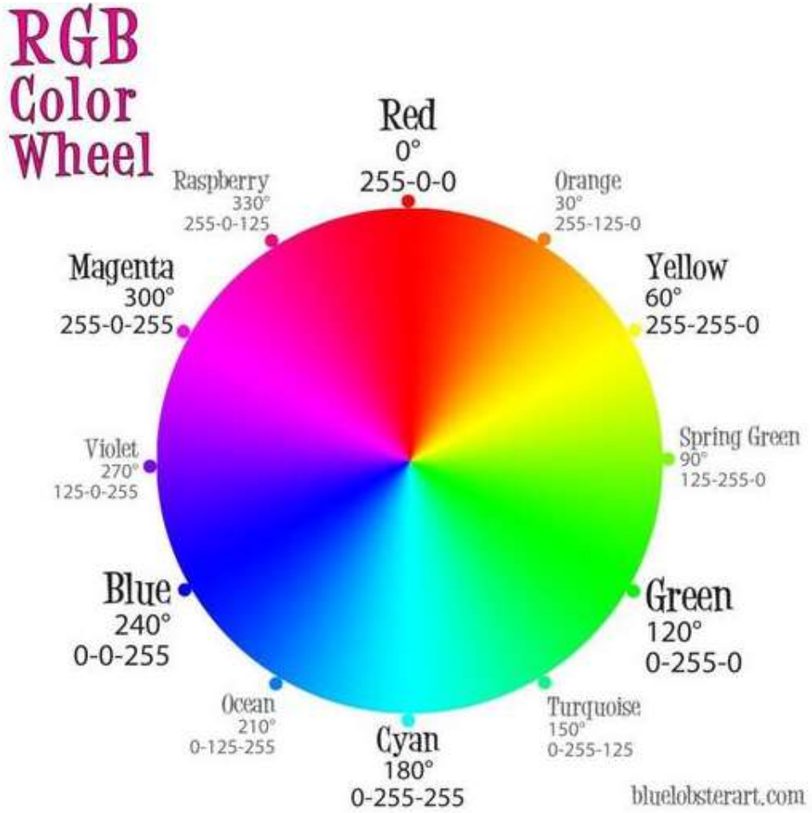





```python
def rgb_wheel(degree):
    degree %= 360
    if 0 <= degree < 60:
        red = 1
        green = degree / 60
        blue = 0
    elif 60 <= degree < 120:
        red = 1 - (degree - 60) / 60
        green = 1
        blue = 0
    elif 120 <= degree < 180:
        red = 0
        green = 1
        blue = (degree - 120) / 60
    elif 180 <= degree < 240:
        red = 0
        green = 1 - (degree - 180) / 60
        blue = 1
    elif 240 <= degree < 300:
        red = (degree - 240) / 60
        green = 0
        blue = 1
    else:  # 300 <= degree <= 360
        red = 1
        green = 0
        blue = 1 - (degree - 300) / 60
    rgb_tuple = (int(red * 255), int(green * 255), int(blue * 255))
    return rgb_tuple
```

ГОРНЫЙ

Цветовой круг




```python
while True:
    for degree in range(0, 361):
        brightness = abs(degree % 60 / 60 * 100 * 2 - 50)
        print(rgb_wheel(degree), brightness)
        switch_on_a_colour_of_the_specified_brightness(rgb_led,
rgb_wheel(degree), brightness)
        sleep(0.05)
```

Python

```python
from machine import PWM
from time import sleep

RED_PIN = 15
GREEN_PIN = 14
BLUE_PIN = 13

red_led = PWM(RED_PIN, freq=1000)
green_led = PWM(GREEN_PIN, freq=1000)
blue_led = PWM(BLUE_PIN, freq=1000)

rgb_led = (red_led, green_led, blue_led)

def get_colour_data_for_pwm(color_8bit_tuple):  …


def set_the_colour_brightness(color_tuple, brightness):  …


def switch_on_a_color(rgb_led, color_tuple):  …


def switch_on_a_colour_of_the_specified_brightness(rgb_led, color_tuple,
brightness=1):  …

def rgb_wheel(degree):  …


while True:
    for degree in range(0, 361):
        color_tuple = get_colour_data_for_pwm(rgb_wheel(degree))
        print(rgb_wheel(degree))
        switch_on_a_color(rgb_led, color_tuple)
        sleep(0.05)
```

ГОРНЫЙ

Контрольные вопросы

- Как правильно подключить «чистый» RGB диод?
- Как можно управлять цветом RGB диода и его яркостью?
- Зачем использовать коллекции и функции?
- Какой синтаксис создания функций?
- Какие требования предъявляются к функциям?
- Что такое принцип DRY?
- Что такое принцип единственной ответственности?
- Что такое декомпозиция?
- Что такое интерфейс?
- Что такое «колесо цвета» и как его описать функцией?